7. Music is been played in two opposite galleries with certain group of people. In the first gallery a group of 4 singers were singing and in the second gallery 9 singers were singing. The two galleries are separated by the distance of 70 m. Where should a person stand for hearing the same intensity of the singers voice? (Hint: The ratio of the sound intensity is equal to the square of the ratio of their corresponding distances).

8. There is a square field whose side is 10 m. A square flower bed is prepared in its centre leaving a gravel path all round the flower bed. The total cost of laying the flower bed and gravelling the path at ₹3 and ₹4 per square metre respectively is ₹364. Find the width of the gravel path.

9. The hypotenuse of a right angled triangle is 25 cm and its perimeter 56 cm. Find the length of the smallest side.

3.6.6 Nature of Roots of a Quadratic Equation

The roots of the quadratic equation $ax^2 + bx + c = 0,\ a \neq 0$ are found using the formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Here, $b^2 - 4ac$ called as the discriminant (which is denoted by Δ) of the quadratic equation, decides the nature of roots as follows

Values of Discriminant $\Delta = b^2 - 4ac$	Nature of Roots
$\Delta > 0$	Real and Unequal roots
$\Delta = 0$	Real and Equal roots
$\Delta < 0$	No Real root

Example 3.40 Determine the nature of roots for the following quadratic equations

(i) $x^2 - x - 20 = 0$ (ii) $9x^2 - 24x + 16 = 0$ (iii) $2x^2 - 2x + 9 = 0$

Solution (i) $x^2 - x - 20 = 0$

Here, $a = 1,\ b = -1,\ c = -20$

Now, $\Delta = b^2 - 4ac$

$\Delta = (-1)^2 - 4(1)(-20) = 81$

Here, $\Delta = 81 > 0$. So, the equation will have real and unequal roots

(ii) $9x^2 - 24x + 16 = 0$

Here, $a = 9,\ b = -24,\ c = 16$

Now, $\Delta = b^2 - 4ac = \left(-24\right)^2 - 4\left(9\right)\left(16\right) = 0$

Here, $\Delta = 0$. So, the equation will have real and equal roots.

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